

COGNITIVE STATE PACKET (CSP): TYPED 9-TUPLE $\langle G, F_t, C_t, E_t, A_t, U_t, D_t, Q_t, O_t \rangle$

Goal G

Immutable Task Prompt

$G \in \Sigma^*$ (Invariant)

Verified Facts F_t

Deduplicated Fact Set

Provenance Tracked

Calculations C_t

(expr, val, unit)

[VERIFIED: AST]

Evidence E_t

(claim, src, c)

Diversity Discount

$w_{\text{div}} = 0.65$

Epistemic: A_t, U_t

A_t : Working Assumptions

U_t : Known Uncertainties

Explicit Epistemic State

Decisions: D_t, Q_t

D_t : Architectural Choices

Q_t : Open Questions

Guards Against Drift

Digital Artifact Map $O_t = \{k_i \mapsto v_i\}$

Deliverables: Python scripts, numerical matrices, simulation data, LaTeX tables

Non-Destructive Versioning: Collision $k \rightarrow k_vN$ with $k_CONFLICT = \text{True}$

Preserves code integrity across specialist handoffs

Trace Audit H_t

$[h_0, \dots, h_{t-1}]$

Model IDs & Timestamps

Latency & Memory Audit

TOKENIZER-INDEPENDENT HANDOFF

Upstream Specialist M_A (e.g., Qwen-Math)

Vocab $\mathcal{V}_A = 152\text{ k}$ | Hidden $\mathbb{R}^{d_A}$ | Emits ΔS_t

Calculates: $\Delta C_t = \{('4 * \pi * 1e - 7', 1.257 \times 10^{-6}, 'H/m')\}$

Independent AST Verifier: $|y_{\text{comp}} - y_{\text{decl}}| \leq \epsilon$

to_prompt_context(): Clean Markdown

VERIFIED CALCULATIONS

- [VERIFIED: ast] mu_0 = 1.2566e-6 H/m

VERIFIED FACTS & ARTIFACTS

Downstream Specialist M_B (e.g., DeepSeek-Coder)

Vocab $\mathcal{V}_B = 32\text{ k}$ | Completely Independent Architecture

Consumes verified context with ZERO amnesia & ZERO drift